# HAOMING ZHANG

Robotics - Mechanical Engineering

## CONTACT


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## RESEARCH INTEREST

**State Estimation**

Kalman filters
Graph Optimization
GNSS & SLAM & Sensor Fusion

**Variational Inference**

Variational Bayes
Parameter Tuning
Online Model Identification

**Deep/Reinforcement Learning**

Object Recognition
Active Perception

## EDUCATION

**Ph. D. - Mechanical Engineering** — 2020 - 2025
**RWTH Aachen University (Germany)**
Topic: Uncertainty Quantification in Factor Graphs for Vehicle Localization

**M. Sc. - Mechatronics** — 2015 - 2016
**University of Duisburg-Essen (Germany)**

**B. Sc. - Mechanical Engineering** — 2012 - 2015
**University of Duisburg-Essen (Germany)**

## WORK EXPERIENCE

**Research Associate** — Jul 20 - Aug. 24
**Institute of Automatic Control, RWTH Aachen Uni.**
Research on state estimation and learning-based inference;
Project: autonomous shipping

**Research Associate** — May 17 - Jun 20
**Cybernetics Lab, RWTH Aachen Uni.**
Research/Teaching: state estimation and trajectory planning;
Project: human/robot collaboration and mobile manipulation

**Student Research Associate** — May 16 - Feb 17
**Chair of Dynamics and Control, Uni. Duisburg-Essen**
Research: simulation design for autonomous driving

## PUBLICATIONS

◇: first author

**GNSS/Multi-Sensor Fusion Using Continuous-Time Factor Graph Optimization for Robust Localization** — ◇ 2024
**IEEE Transactions on Robotics**
DOI: 10.1109/TRO.2024.3443699

**Learning-based NLOS Detection and Uncertainty Prediction of GNSS Observations with Transformer-Enhanced LSTM Network** — ◇ 2023
**Proc. of IEEE 26th Int. Conf. on Intelli. Trans. Syst.**
DOI: 10.1109/ITSC57777.2023.10422672

**Object Detection and Heading Estimation from Radar Raw data** — 2022
**Proc. of 34th IEEE Intelligent Vehicles Symposium**
DOI: 10.1109/IV55152.2023.10186591

**Continuous-Time Factor Graph Optimization for Trajectory Smoothness of GNSS/INS Navigation in Temporarily GNSS-Denied Environments** — ◇ 2022
**IEEE Robotics and Automation Letters**
DOI: 10.1109/LRA.2022.3189824

# SKILLS

**C++ & Python** 7+ yrs

**ROS** 7+ yrs

**Machine Learning** 5+ yrs

**Teaching** 3+ yrs

**Web Development** <1 yrs

# LANGUAGES

**English**
Proficient (C1)

**German**
Fluent (C2)

**Chinese**
Native

**Predicting basin stability of power grids using graph neural networks** 2022
New Journal of Physics
DOI: 10.1088/1367-2630/ac54c9

# PROJECTS

**MuSEAS: multi-sensor dataset with annotations for ship automation** 2023-2024
**Role: Project Leader**
Visual odometry; Multi-sensor fusion; Mapping; Hardware/Software concept; Project management

**FernBin: Remote-controlled and coordinated driving in land shipping** 2020-2024
**Role: Project Leader**
Robust ship localization; Sensor fusion; Hardware/Software concept; Project management

**UrbANT: urban, automated, user-oriented transport platform** 2019-2020
**Role: Project Collaborator**
Outdoor Robot localization; Object detection/tracking; Robot design and prototyping; Project management

**ARIZ: work in the industry of the future** 2017-2019
**Role: Project Collaborator**
Indoor robot localization/SLAM;
Trajectory planning of mobile/stationary robots

# ACTIVITIES

**3nd Workshop on Intelligent Vehicle Meets Urban: Safe and Certifiable Navigation and Control for Intelligent Vehicles in Complex Urban Scenarios** Edmonton, CA
**Role: Co-Organizer and Moderator**
More Information

**2nd Workshop on Intelligent Vehicle Meets Urban: Safe and Certifiable Navigation and Control for Intelligent Vehicles in Complex Urban Scenarios** Bilbao, ES
**Role: Co-Organizer and Moderator**
More Information

**Demonstrations of 33rd IEEE Intelligent Vehicles Symposium** Aachen, DE
**Role: Organizer**
More Information

**Talk: Closing the Estimation Loop: Learning-based Uncertainty Quantification for Robust Vehicle Localization** Beijing, CN
**Role: Presenter**
Tsinghua University, 2024

**Talk: Towards Robust Navigation Solution and Flexible Sensor Fusion in Challenging Inland Shipping Scenarios**

**Duisburg, DE**

**Role: Presenter**

The Autonomous Inland and Short Sea Shipping Conference, 2022

**Reviewer for:**

**AoE**

IEEE T-RO/RA-L/T-AES/T-WC/Sensors
IEEE ICRA/IROS

**References**

**AoE**

Prof. Dr. Heike Vallery (h.vallery@irt.rwth-aachen.de)
Prof. Dr. Timothy D. Barfoot (tim.barfoot@utoronto.ca)
Prof. Dr. Dirk Abel (d.abel@irt.rwth-aachen.de)
Prof. Dr. Li-Ta Hsu (lt.hsu@polyu.edu.hk)